

Dear Editor,

We are pleased to submit our manuscript entitled “Congestion-Aware Joint Charging-Station Placement, Capacity, and Swarm-UAV Trajectory Design for Age-Optimal Data Collection” for consideration for publication in the *IEEE Internet of Things Journal*.

Unmanned aerial vehicles increasingly collect time-sensitive Internet of Things data, where the quality metric is the Age of Information. Practical deployments use a swarm of energy-limited UAVs that recharge at a few shared charging stations, which then become a contended resource whose waiting time inflates the Age of Information. Jointly deciding where to place the stations, how many charging ports to install, how the swarm should fly, and which UAV recharges where is a coupled problem that, to our knowledge, has not been addressed as a whole.

Our contributions are: (i) the first joint formulation of charging-station placement, port capacity, swarm trajectory, and assignment for peak-Age-of-Information data collection with a contended, multi-server charging resource; (ii) a finite-source (machine-repair) queue model, validated against a discrete-event simulation, which shows that the commonly assumed open $M/M/c$ queue overpredicts the waiting time by roughly six times and would distort the design; (iii) two proved structural results, a congestion threshold that makes the Age of Information non-monotone in the swarm size with a provisioning law, and a placement inversion from coverage-driven to traffic-driven; and (iv) a block-coordinate solver with a GPU close-enough traveling-salesman trajectory optimizer, evaluated against several baselines with large-swarm scaling, robustness, and optimality-gap studies.

The topic aligns with the scope of the journal and will interest its readership working on UAV-assisted data collection, Age of Information, and energy-aware Internet of Things systems.

We confirm that the manuscript is original, has not been published previously, and is not under consideration elsewhere; all authors approved the submission and declare no conflict of interest. To support reproducibility, the source code and data are available at <https://github.com/haodpsut/swarm-uav-aoi-congestion>.

Thank you for considering our manuscript.

Sincerely,

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